

# Obesity

- Definition,
- Causes,
- Hormones,
- Assessment,
- Complications
- Conclusion

# **Definition of Obesity**

- A medical condition characterized by excessive body fat accumulation
- Typically defined by Body Mass Index (BMI)  $\geq 30 \text{ kg/m}^2$  indicates obesity

# **Causes of Obesity**

Obesity results when energy intake exceeds energy expenditure. This imbalance results from a complex interaction of biochemical, neurologic, environmental and psychologic factors.

## **Risk factors:**

- Genetic factors
- Environmental factors (diet, lifestyle)
- Psychological factors
- Sedentary behavior
- Socioeconomic influences

# Hormones Involved in Obesity

- • Leptin – regulates satiety
- • Ghrelin – stimulates hunger
- • Insulin – regulates glucose metabolism
- • Cortisol – stress hormone linked to fat accumulation
- • Thyroid hormones – regulate metabolism

# Assessment of Obesity

- • Body Mass Index (BMI)
- • Waist circumference
- • Waist-to-hip ratio
- • Skinfold thickness
- • Clinical evaluation and history

# **Body mass index (BMI)**

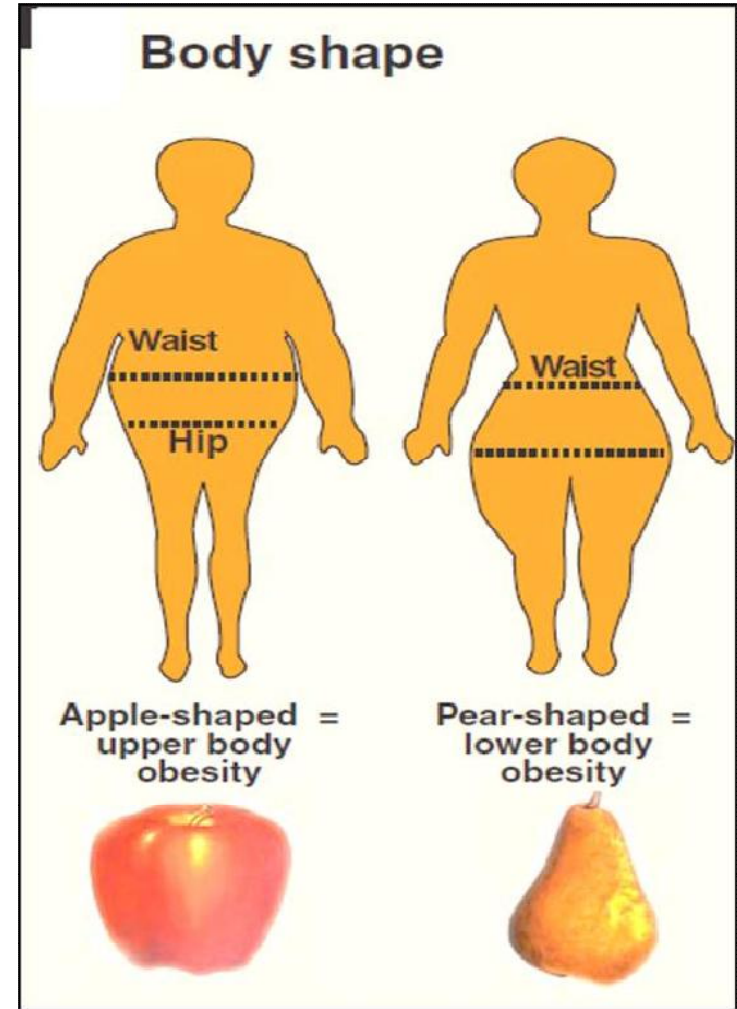
which provides a measure of relative weight, adjusted to height. The healthy range of BMI is 18.5 to 24.9 kg/m<sup>2</sup>.

## **BMI Classification**

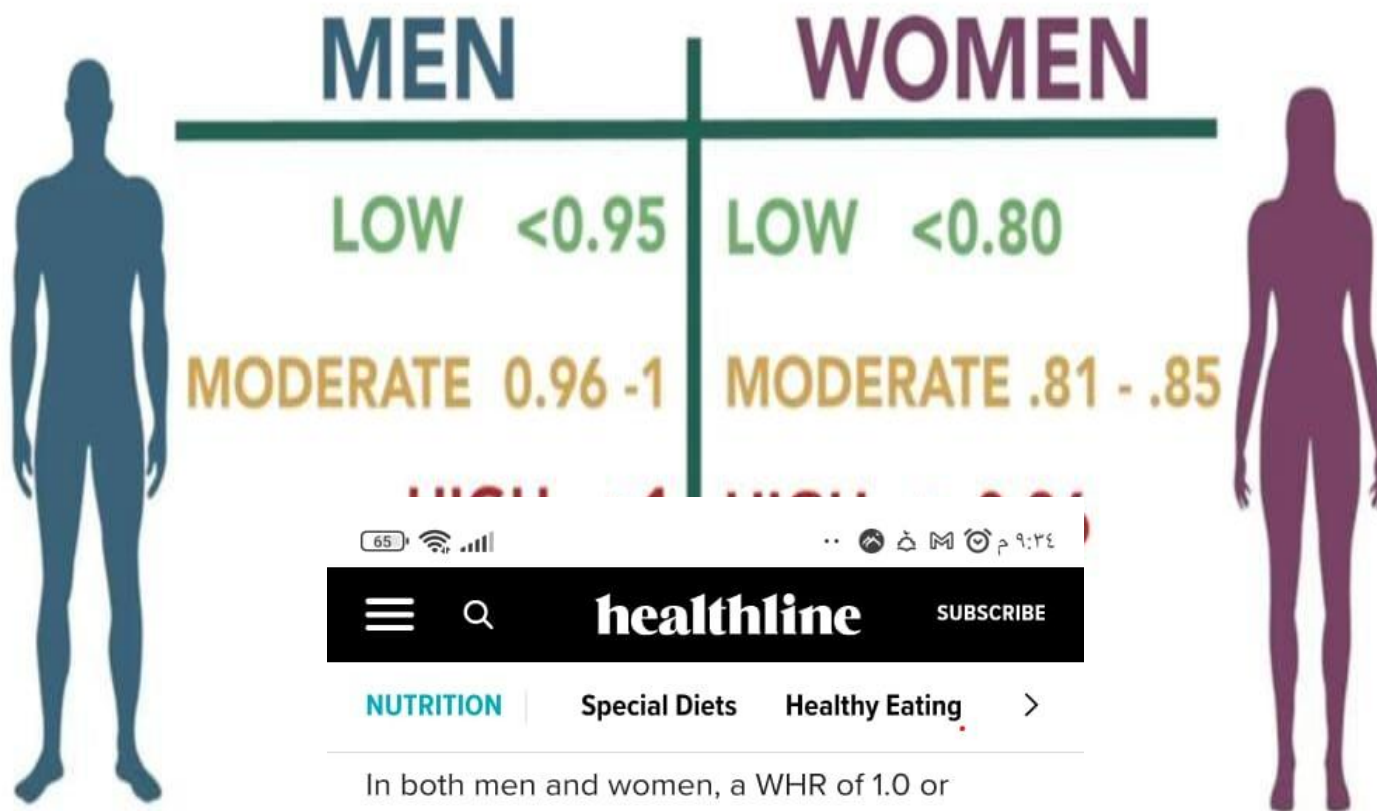
- Underweight: <18.5
- • Normal: 18.5–24.9
- • Overweight: 25–29.9
- • Obesity: ≥30

# Waist to hip ratio

- A waist to hip ratio of more than 0.8 for women and more than 0.9 for men is defined as 'apple-shaped' which is associated with more fat deposition in the trunk.
- In contrast, a lower waist to hip ratio reflects a preponderance of fat distributed in the hips and thighs and is called 'pear-shaped'



THE WORLD HEALTH ORGANIZATION CLASSIFIES  
HEALTH RISK WITH THE FOLLOWING INFORMATION:



Waist-to-hip ratio chart



# Complications of Obesity

- Type 2 diabetes mellitus
- Cardiovascular diseases
- Hypertension
- Osteoarthritis
- Sleep apnea
- Certain cancers

# **Conclusion**

- Obesity is a multifactorial condition
- Requires lifestyle, medical, and sometimes surgical management
- Early assessment helps prevent complications